

RECORD LAYOUT CHART FOR SCHEDULED COURSE FILE name of file

Field Name	SECTION NUMBER	COURSE NUMBER	ROOM NUMBER
Characteristics*	ALPHA STRING	ALPHA STRING	ALPHA STRING
Position**	0 1	3 4	6 7

DAYS SCHEDULED	INSTRUCTOR LAST NAME	INSTRUCTOR FIRST
ALPHA STRING	ALPHA STRING	ALPHA STRING
20 33	34 63	64

**POSITION
Hexadecimal / Decimal
Numbering
from
00 to FF / 0 to 255

HEXADECIMAL DECIMAL	00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
HEX DEC	00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25
HEX DEC	40 41 42 43 44 45 46 47 48 49 4A 4B 4C 4D 4E 4F 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89
HEX DEC	80 81 82 83 84 85 86 87 88 89 8A 8B 8C 8D 8E 8F 90 91 92 93 94 95 96 97 98 99 A0 A1 A2 A3 A4 A5 A6 A7 A8 A9 AA AB AC AD AE AF B0 B1 B2 B3 B4 B5 B6 B7 B8 B9 BA BB BC BD BE BF C0 C1 C2 C3 C4 C5 C6 C7 C8 C9 CA CB CC CD CE CF D0 D1 D2 D3 D4 D5 D6 D7 D8 D9
HEX DEC	C0 C1 C2 C3 C4 C5 C6 C7 C8 C9 CA CB CC CD CE CF D0 D1 D2 D3 D4 D5 D6 D7 D8 D9 E0 E1 E2 E3 E4 E5 E6 E7 E8 E9 EA EB EC ED EE EF F0 F1 F2 F3 F4 F5 F6 F7 F8 F9

NAME	
Gr	
78	

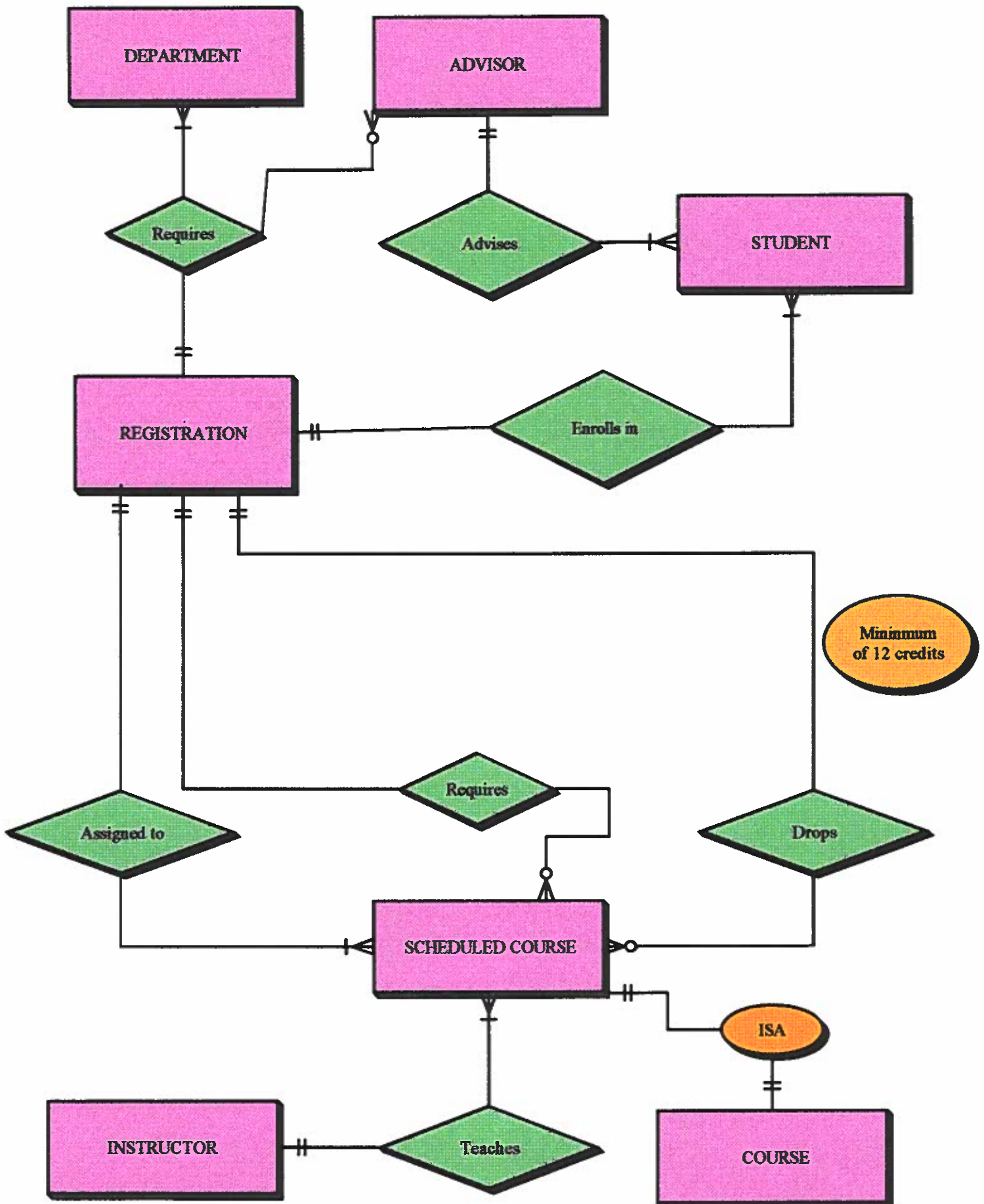
Notes on Physical Database Schema

- 1) The Schema is implemented with the Data Definition Language, SQL.
- 2) The DBMS used for the Physical Database Schema is Oracle.
- 3) All primary keys have key, domain and referential integrity.
- 4) The rules for deletion of tables are given near the tables in the Schema.

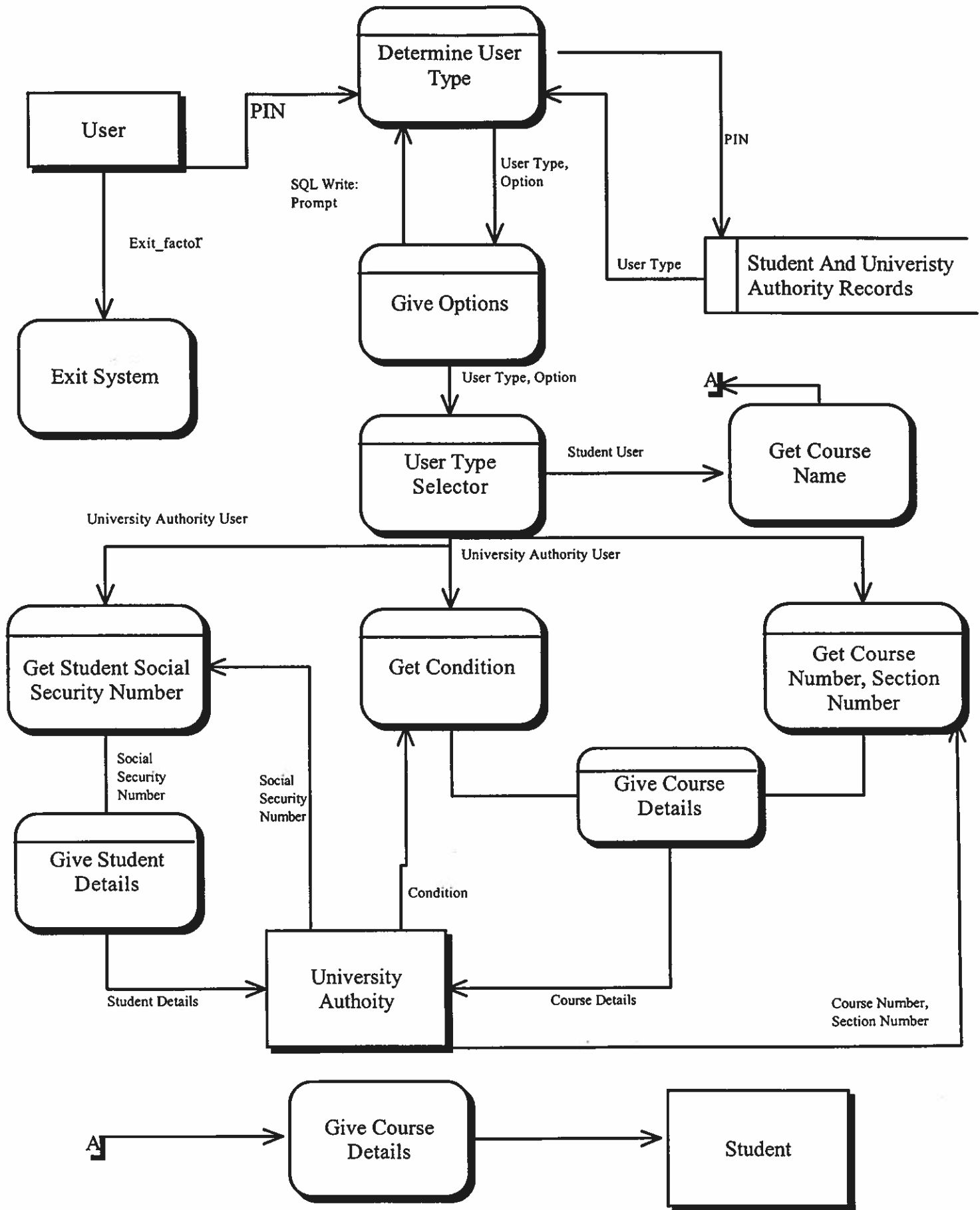
They denote:

- | | | |
|----|-------------------|------|
| a) | No restriction: | D:N |
| b) | Delete Cascade: | D:C |
| c) | Delete Restrict: | D:R |
| d) | Delete: Set Null: | D:SN |

NORMALIZED ERD (All attributes are listed on the next page)



Information Retrieval *Physical DFD*

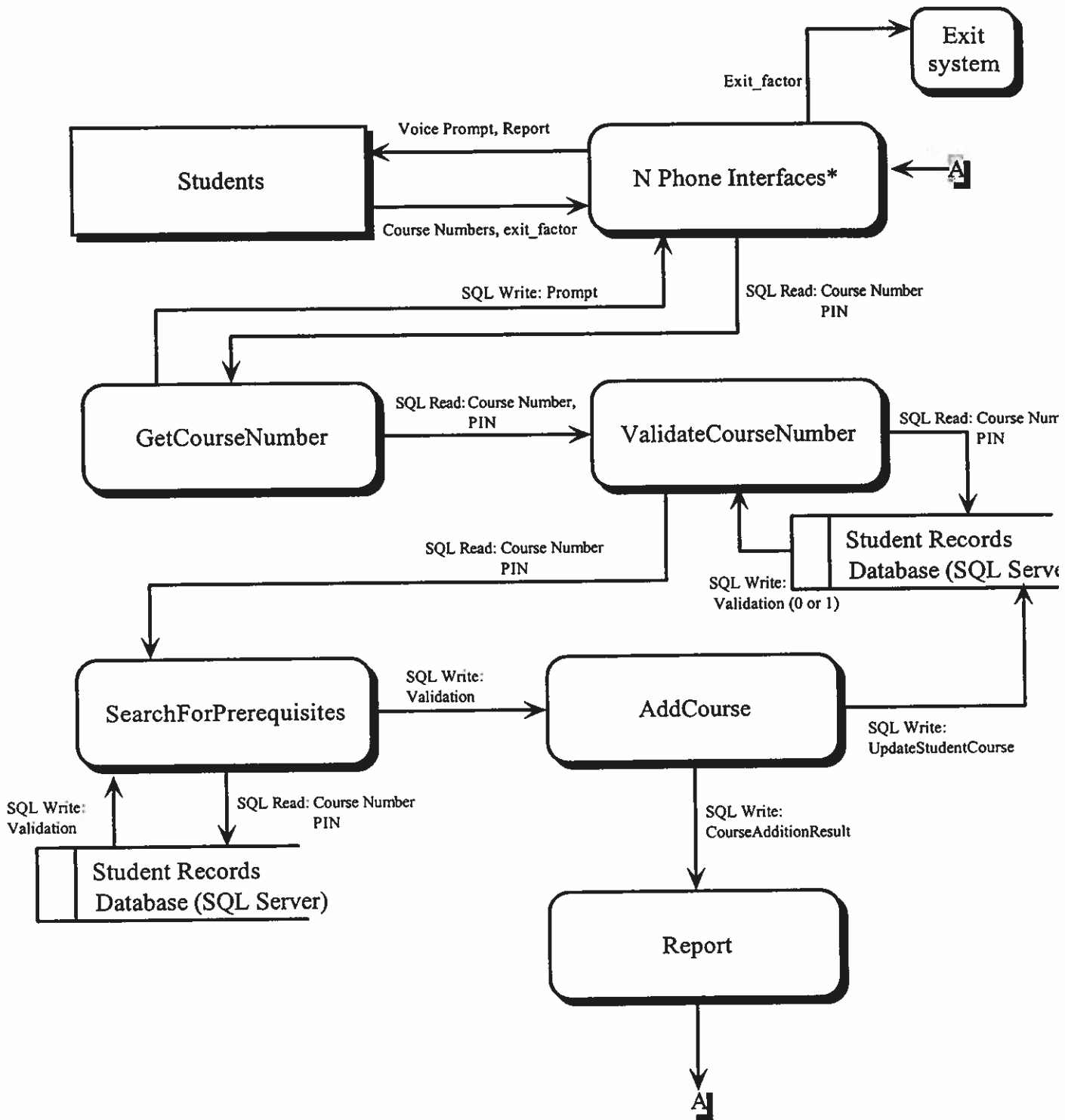


Note: The Process "Give Course Details" utilizes the Course List Database on the University Serv
Also, all data flows are in SQL.

Brief Narrative of Information Retrieval

In the diagram, the User enters the User's Pin and the system recognizes the User as a Student or an University Authority. After the system recognizes the User, the Voice Prompts the user's options for that type of User. The Student gets the name of a Course and all the information about that Course. If the User is an University Authority, the official can enter the Student's Social Security Number and retrieve all the necessary information about that Student, or the official can enter the Course Number and Section to retrieve all the necessary information about that Course.

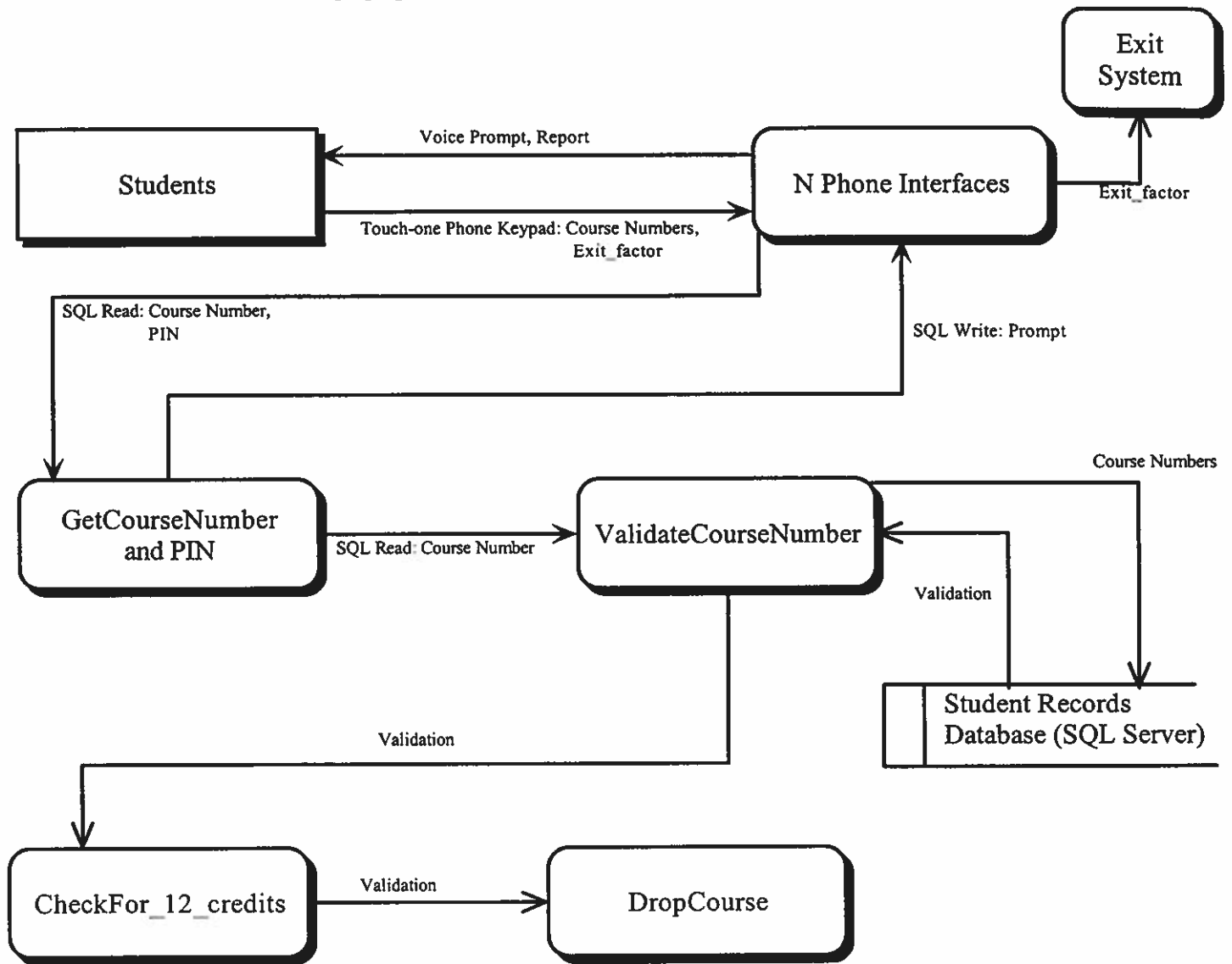
COURSE ADDITION *PHYSICAL* DFD



Brief Narrative of Course Addition

The diagram shows Students (denoted by the square box) adding a course. The data flow associated with Students are, Voice Prompt, Report and Course Numbers. The Voice Prompt guides Students through the process of adding a course. So it is shown as an output. Report is generated when the system has either added the course or not. It tells the student whether the course was added. It also gives reasons for not adding a course. The Students input Course Numbers of the courses they want to register for. Next in the DFD is the Phone interface; it consists of touch-tone which the Students use, and the AT&T communication Server which connects the Oracle Server and phone lines. It acts as interface to the University LAN and phones. The phone interfaces pass on the Students' PIN and the Course Numbers entered, to the Oracle Server. The server has a set of programs, which add courses. The first of these is GetCourseNumber. This is asks the student for the Course Numbers of the courses to be added. After receiving the Course Numbers, it passes them on to ValidateCourseNumber, which checks to see if the student entered the correct Course Number, against Student Records. After validation, the program SearchForPrerequisites checks whether the student has the necessary prerequisite courses. If so, the program AddCourse adds the courses to the Student's list of courses.

COURSE DROP *PHYSICAL* DFD



Brief Narrative of Course Drop Subsystem

In the diagram, the Student uses a phone to key in the Course Number of the courses they want to drop. Student is guided by the Voice Prompt. The data is then transmitted over phone lines to the logic based in the University Oracle Server. This set of logic consists of the following programs: GetCourseNumber, ValidateCourseNumber, Check_For_12_Credits, and DropCourse. GetCourseNumber prompts the Student for Course Number and Pin, and it passes on the information to ValidateCourseNumber, which checks if the Course Number input is valid, by checking it against the database of Courses. If valid, the program Check_For_12_Credits calculates to see whether the Student will have 12 credits after student drops the course. If so, the program drops the course from the Student's List of courses. If any of these programs don't find a valid reason to proceed, they abort the Course Drop and inform the Student about reasons for doing so. An automatic backup is created when the updates take place.

Legend For System Flowcharts



An Online Input



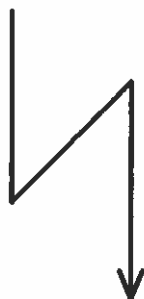
A Program



An Online Output

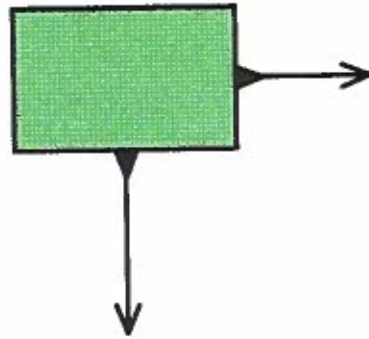


A Disk Drive



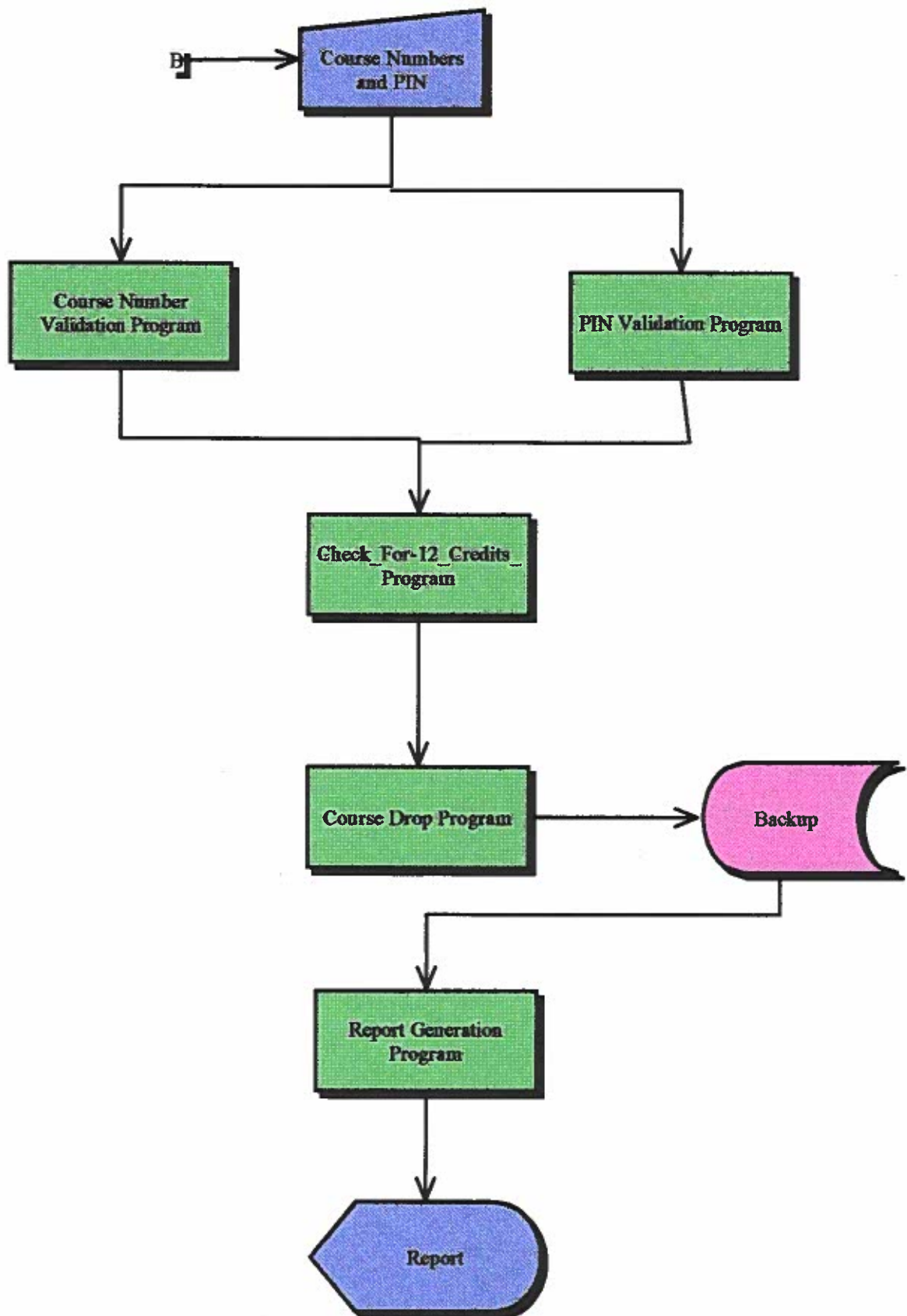
Communication/Data Upload/Download

Legend cont...

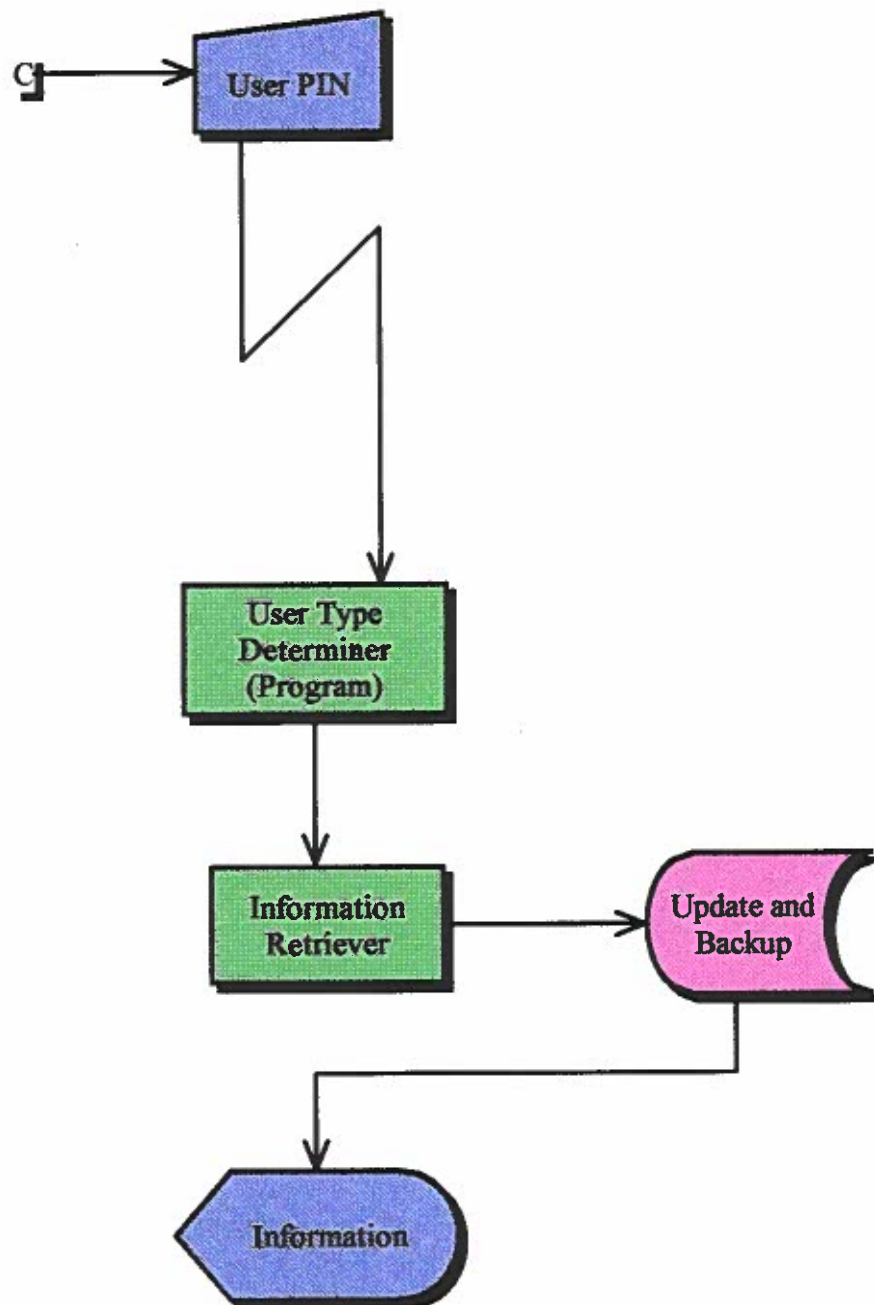


Denotes Either One Or Other Sequence

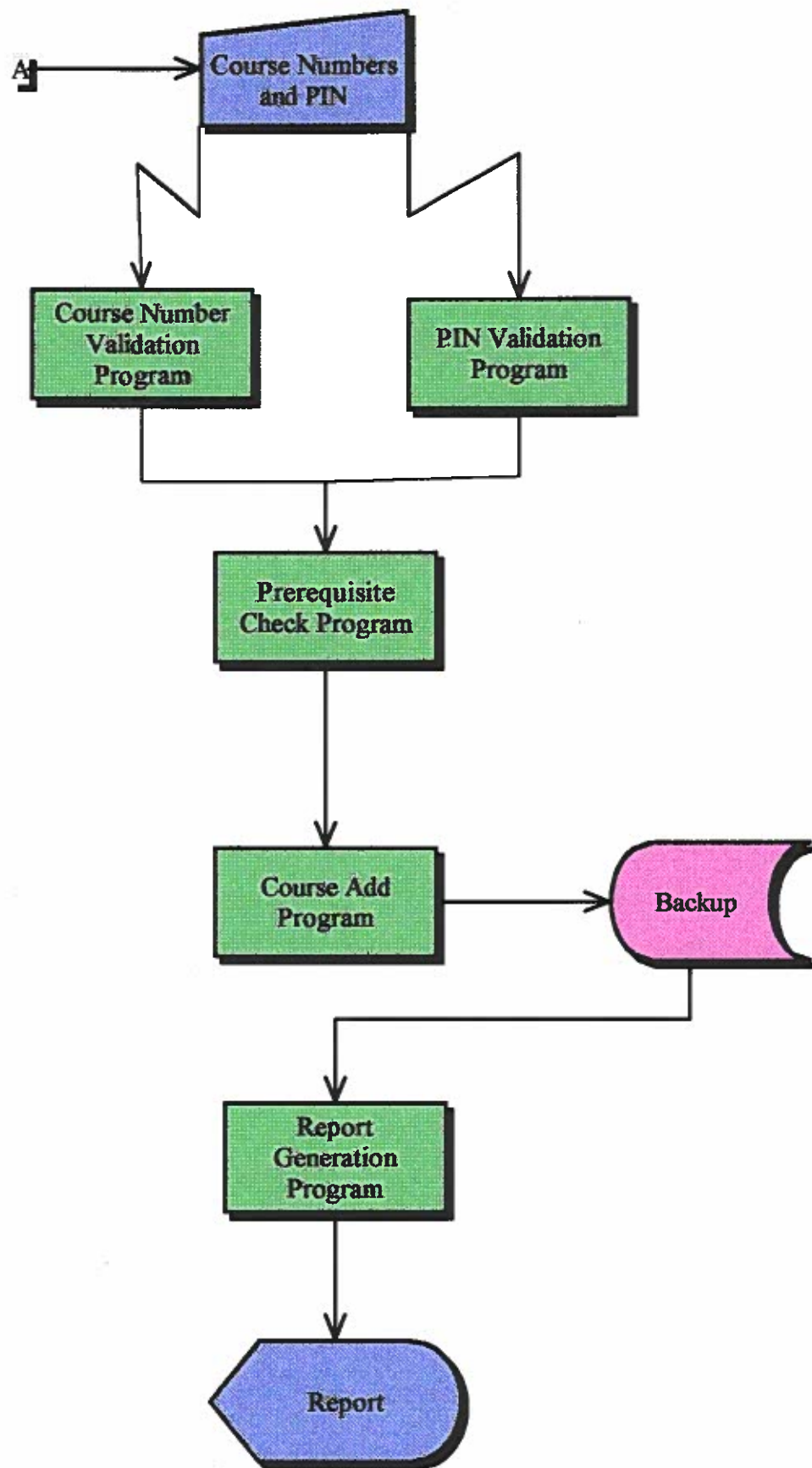
Course Drop System Flowchart



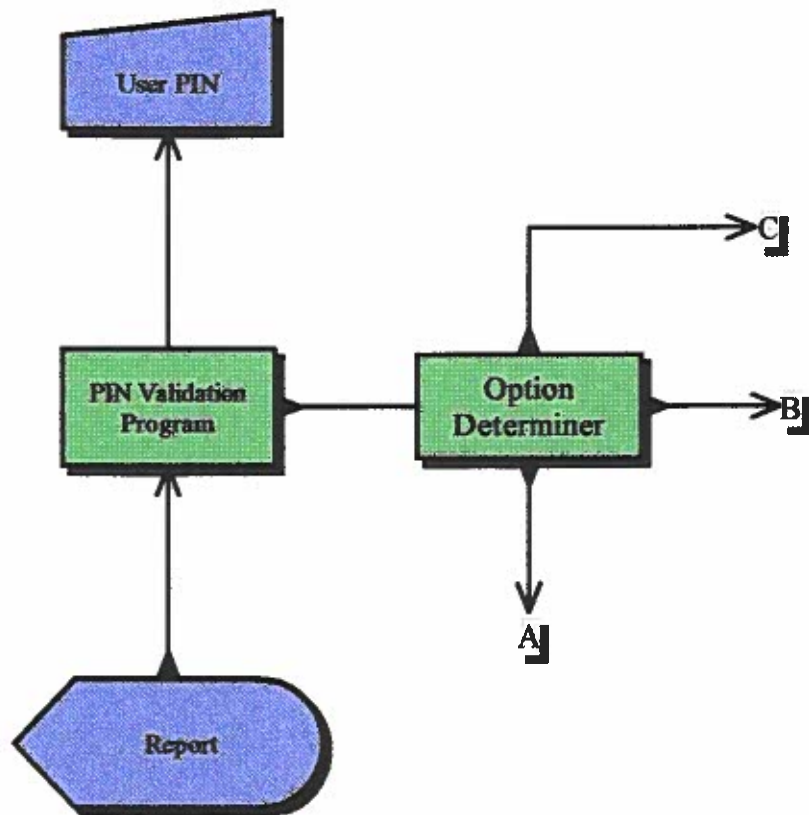
Information Retrieval



Course Addition System Flowchart



Security Check System Flowchart



Brief Narrative of Terminal Report Chart for Exception Report on Students

This terminal display chart gives an exception report on students. This report filters data before they are presented to the manager as information. This report reports only the student that has exception to some condition or standard; an example of an exception report is an report that list all information about a student that owe a balance on their tuition, or the student may be on probation due to grades or their behavior at the university.

Brief Narrative Of Terminal Display Chart for Summary Report On Scheduled Courses

This terminal display chart gives a summary of scheduled courses. It doesn't include all the details of the course; only the attributes that the user has requested. In the case of this chart, the attributes are Course Number, Section Number, Course Title/Name, Start Time and End Time.

PRINTER SPACING CHART FOR DETAIL REPORT ON STUDENTS
Name of Report/Document

PREPARED BY _____ DATE _____

PROGRAM NAME _____

CARRIAGE CONTROL

[illegible]

Brief Narrative of Terminal Report Chart for Detailed Report on Students

This terminal display chart gives and detailed report on the students attending the university. The detailed report on students presents information with little on filtering or restrictions. This report lists all information about the student along with their pin.

Brief Narrative Of Terminal Display Chart for Summary Report On Students

The Terminal Display Chart shows a brief summary of students' names, their PINs, social security numbers, whether they were graduates or undergraduates (Student Type) and total number of credits each has earned.

Brief Narrative Of Terminal Display Chart for Detailed Report On Courses

This chart gives all the details of courses. It gives the Course Number, Course Name/Title, Number of Credits that can be earned and the Number of the department that offers the course. Please note that it doesn't give the details of the sections and the particulars. For that, we would have to search for Scheduled Courses.

Course and Section Number Input

Enter the Course and Section Number of the Course you want to retrieve

Enter Course, Section Number

Course and Section Number Input

Student PIN Input

Enter PIN of Students whose record you want to retrieve

Enter Student PIN

Student PIN Input

Input Screen of Second Menu for University Authorities

The screenshot shows a window titled "Student Records - Menu 2" with standard Windows window controls (minimize, maximize, close). The window contains a list of three menu items, each preceded by a small circular icon:

- Exception Report on Students and Update
- Detailed Report on Students and Update
- Summary Report on Students and Update

At the bottom of the window, there are two buttons: "Enter" on the left and "Cancel" on the right.

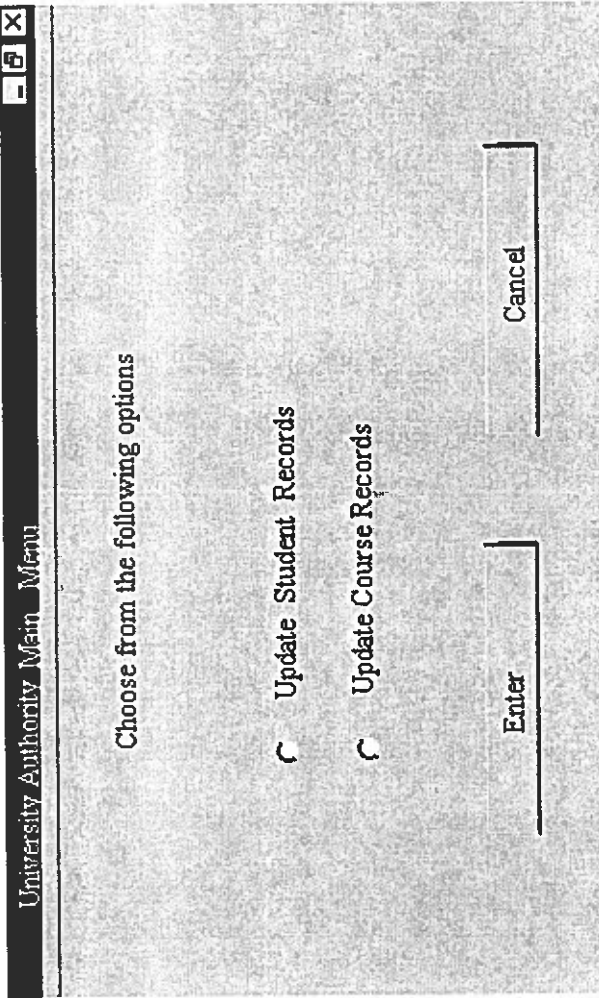
Input Screen of Third Menu for University Authorities

Course Records - Menu 3

- ☐ Exception Report on Courses and Update
- ☐ Detailed Report on Courses and Update
- ☐ Summary Report on Courses and Update

Enter Cancel

Input Screen of Main Menu for University Authorities



The screenshot shows a graphical user interface window titled "University Authority Main Menu". The window has a standard title bar with minimize, maximize, and close buttons. The main content area contains the instruction "Choose from the following options" followed by two radio button options: "Update Student Records" and "Update Course Records". At the bottom of the window, there are two buttons: "Enter" on the left and "Cancel" on the right.

University Authority Main Menu

Choose from the following options

☐ Update Student Records

☐ Update Course Records

Enter Cancel

Input Screen for PIN, to access Registration System

The image shows a screenshot of a software window titled "Access Validator". The window has a standard Windows-style title bar with minimize, maximize, and close buttons. The main area of the window has a textured, gray background. In the center, the text "Enter PIN" is displayed above a white rectangular input field. The text "U23Z" is entered into this field. Below the input field, there are two buttons: "Enter" on the left and "Cancel" on the right. Both buttons are rectangular with a thin black border and a light gray fill.

Access Validator

Enter PIN U23Z

Enter Cancel

Detail Report on Students

Detail Report - Student	
William Kahan	
PIN:	U23Z
Social:	[REDACTED]
Type:	GRAD
Term:	SPR97
Race:	Hispanic
Major:	CompSci
Minor:	None
Sex:	M
DOB:	17-07-77
Temp Add:	12, northwood drive, jamaica,ny,ny 11010
Perm Add:	12, northwood drive,
Exit to Main Menu	
Scroll down to see more	